SK Saif Ibna Ezhar Arko



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LinkedIn — GitHub — Portfolio

Research Interests

Quantum Information Theory • Quantum Computing and Algorithms • Quantum Cryptography • Quantum Machine Learning • Adversarial Robustness of Quantum Models • Post-Quantum Cryptography.

Education

BRAC University, Dhaka, Bangladesh *2026*
B.Sc. in Computer Science and Engineering — CGPA: **3.20 / 4.00** (Approx. 78.33% equivalent based on grade-boundary interpolation)

- **Undergraduate Thesis:** *Quantum-Aware Image Encoding and Adversarial Perturbation*
- **Relevant Coursework:** Quantum Computing, Quantum Error Correction, Quantum Algorithms, Machine Learning, Pattern Recognition, Data Structures & Algorithms, Linear Algebra, Probability & Statistics.

Notre Dame College, Dhaka *2020*
Higher Secondary Certificate (Science) — GPA: **5.00 / 5.00**

St. Gregory's High School and College, Dhaka *2018*
Secondary School Certificate (Science) — GPA: **5.00 / 5.00**

Research Experience

Undergraduate Thesis Researcher — BRAC University, Dept. of CSE *2024 – Present*
Supervisor: Ipshita Bonhi Upoma, Lecturer, Department of CSE, BRAC University

- Developing theoretical frameworks for encoding classical image data into quantum states via amplitude and basis encoding, and analyzing their robustness against adversarial noise.
- Studying trade-offs between encoding fidelity, circuit depth, and resilience to bit-flip and phase-flip channels.

Student Researcher — ELITE Research Lab LLC *Sept. 2026 – Jan. 2027*
Supervisor: Md Kishor Morol

- Conducting research at the intersection of quantum computing and machine learning in collaboration with industry experts.

Research Intern, QIntern 2026 Global Internship — QWorld *Summer 2026*
Supervisor: Dr. Maha A. Metawei

- Project: *Accelerating hybrid quantum-classical machine learning tasks on HPC platforms (QNLP).*
- Implemented high-performance quantum circuit simulations and tensor network optimizations to scale hybrid algorithms on distributed clusters; extended core framework classes (*Ansatz*, *Reader*) and built unified data format conversion scripts.

Research Intern, QIntern 2025 Global Internship — QWorld *Summer 2025*
Supervisors: Dr. Kumar Prateek, Dr. Soumyadev Maity

- Project: *Design of Quantum-Anamorphic Encryption Against Coercive Surveillance.* Prototyped anamorphic quantum encryption schemes providing deniability against coercive adversaries.
- **Awarded 3rd Best Project globally** among submissions worldwide; collaborated with an international team across multiple countries.

Professional Experience

Meridial Marketplace, by Invisible, Remote — *AI Trainer (Part-time)* *October 2026 – Present*

- Evaluate and improve large language model outputs to produce high-quality human feedback data for AI training.

Mercor, San Francisco, California, United States (Remote) — *AI Trainer (Part-time)* *October 2026 – Present*

Datasoft Systems Bangladesh Limited, Dhaka *December 2025 – Present*
Software Developer (AI and Quantum Department)

- Deploy and operate self-hosted LLM inference infrastructure with `llama.cpp` and `llama-server`, serving a 228B-parameter (10B active) Mixture-of-Experts model in quantized GGUF format across heterogeneous single-GPU nodes.
- Designed the *Hermes Agent* framework: model-switching, multi-provider LLM routing (NVIDIA NIM, DeepSeek, OpenRouter, local Ollama) behind an authenticated API gateway running as a supervised `systemd` service.

- Built LangGraph-based agentic SQA workflows executing multi-step verification across Django/React/PostgreSQL projects, and delivered *iSAGE*, an internal assistant UI over the local LLM backend.
- Engineer production Python systems for document processing and workflow automation, including distributed scraping and ETL pipelines; contributed to the terminal operating system of Chittagong Sea Port.

Maxcode Lab, Dhaka — *Python Developer Intern* *October 2025 – December 2025*

- Engineered Python tools for web scraping, automation, database management, and REST API integration.

Selected Projects

QVTM — Quantum Variable Topic Model (Global Quantum Hackathon 2024) [GitHub]

- Built an encoder–decoder variational quantum model using **amplitude embedding** to represent text in Hilbert space, optimized with a composite **KL-divergence**, **Cross-Entropy**, and **MMD** objective.
- Achieved a **Topic Diversity score of 1.0** with improved coherence over Latent Dirichlet Allocation baselines.

Monkeypox Classification Using a Hybrid Quantum CNN (HQCNN) [GitHub]

- Combined classical feature extraction with parameterized quantum circuits and benchmarked against purely classical CNN and EfficientNetB3 transfer-learning baselines for medical image diagnosis.

Lexora — Bengali-First English Vocabulary Learning Platform [Live] [GitHub]

- Built a Django 5 and HTMX progressive web app (˜9,900 lines of Python across five apps, 89 tests) for GRE and IELTS preparation, serving 1,468 words in 62 sequential packs with paired Bangla and English meanings.
- Implemented a full **SuperMemo-2** spaced-repetition scheduler alongside a Leitner five-box ladder and an **Ebbinghaus retention-curve** exam forecast, plus study modes spanning timed drills, dictation, confusable-pair training, and an AI-driven debate arena.

NLP Similarity Classifier — Bidirectional LSTM Siamese Network [GitHub]

- Designed a TensorFlow/Keras Siamese network for paraphrase identification, fusing sentence embeddings with Jaccard and fuzzy-matching features across four merge strategies; an augmentation pipeline generated **7,400+ synthetic samples**, evaluated via ROC curves and confusion matrices.

Other projects: Stock price forecasting — ARIMA vs. LSTM; SmartCare, a Django healthcare management platform; 3D Ball Roller, a physics-based OpenGL game in Python.

Community Contributions, Teaching, and Professional Service

IBM Qiskit Advocate (#129943) — IBM Quantum *2025 – Present*

- Recognized for sustained contributions to the global quantum community; lead workshops, mentor early-career researchers on Qiskit projects, create tutorials and learning resources, support the community on official Discord and GitHub channels, and organize outreach events and hackathons.

QBangladesh — National Hub of QWorld, Bangladesh *April 2025 – Present*
Founder, Coordinator, and Mentor

- Established and lead the first national quantum computing community in Bangladesh under the QWorld umbrella, building a sustainable ecosystem for quantum education and research.
- Deliver hands-on workshops on quantum computing with Python, Qiskit, and Cirq for students nationwide, and mentor early-career researchers on quantum algorithms and simulation.

Caretutors Technology Limited, Dhaka — *Verified Python and Computer Science Tutor December 2024 – Present*

- Teach Python programming to students with varied backgrounds, covering programming logic, core language concepts, data structures, and algorithmic thinking through personalized, project-based curricula.
- Conduct individual and small-group lessons, assess assignments and tests, provide structured feedback, and monitor each learner's progress toward their academic and programming goals.

Technical Skills

Programming:	Python, C, C++, SQL
Quantum:	Qiskit, Cirq, PennyLane (basic), Stim
ML / DL:	TensorFlow, Keras, scikit-learn, NumPy, Pandas, Matplotlib, LangGraph
Web & Tools:	Django, HTMX, PostgreSQL, HTML/CSS, OpenGL, Git, Linux, Jupyter, LaTeX
Languages:	Bengali (Native), English (Full Professional Proficiency)

Talks and Presentations

Invited Talk (Remote): "Quantum Production Scheduling: An Exact QUBO for MetalTech's Three-Machine Shop, Solved with QAOA and QUBO Solvers and Verified against CP-SAT," 83rd Warsaw Quantum Computing Group Meetup, Deep Tech Summit & CEE Challenge, Warsaw, Poland (October 2026).

Honors, Awards, and Certifications

Awards: Best Use of the VeloxQ Platform, "Qubit-Efficient QUBO and XY-QAOA for Shop Scheduling" (solo participation as StegoQubit), QPoland Quantum Hackathon 2026 (QPoland, October 2026); Best Project (2nd Place), "Accelerating Hybrid Quantum-Classical Machine Learning Tasks on HPC Platforms," QIntern 2026 (QWorld, September 2026); Outstanding Participation Diploma, QIntern 2026 (QWorld); 3rd Best Project Globally, QIntern 2025 (QWorld); Diploma in Quantum Anamorphic Encryption, QWorld 2025; Full Financial Support (€1,500) to present a research abstract and attend QUIKSTART X Winter School 2025, RPTU Kaiserslautern-Landau, Germany (Quantum Initiative of Rhineland-Palatinate, QUIP).

Certifications: ICTP Physics Without Frontiers: Bangladesh School on Mathematics for Physics 1 — Operators on Hilbert Spaces (ICTP, 2026); Quantum AI Lab — Quantum Cryptography (Bibliotheca Alexandrina, 2026); Quantum Business Foundations (IBM, 2026); Basics of Quantum Information (IBM, 2026); Pan-African Quantum Challenge 2025 (Africa Quantum Consortium, 2026); Quantum Random Number Generation (Quantum Security Defence, 2026); Introduction to Quantum Computing (Qubit by Qubit, 2025); Qiskit Global Summer School 2026 — Quantum Excellence (IBM, 2026); QCOURSE 501-2: Elements of Quantum Computing and Programming (QWorld, 2025); QSilver26 and QBronze144 Workshops (QWorld, 2024); Quantum Machine Learning Fundamentals (Ingenii, 2024).

References

Ipshita Bonhi Upoma
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